

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). At the bottom, a Windows taskbar is visible with several application icons including Chrome, Photoshop, and others. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

MaskedTextBox



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# Definition

## MaskedTextBox

-  A Windows Forms control that restricts user input according to a predefined mask.
-  A custom mask defines the type, position, and format of allowed characters.

# Explanation

## Mask Property

-  The Mask property defines the required input format.

-  Example:

`MaskedTextBox1.Mask = "00/00/0000";`

## Mask Characters

-  `0` → required digit from 0 to 9.

-  `9` → optional digit or space.

-  `L` → required letter.

-  `?` → optional letter.

-  `A` → required alphanumeric character.

-  `a` → optional alphanumeric character.

## Literals

-  Characters such as `/`, `-`, and `:` are displayed as fixed literals.

-  Literals cannot be changed or deleted by the user.

## Procedure

1. Add a MaskedTextBox to the Windows Form.
2. Set the Mask property.
3. Define the required input format.
4. Use mask characters for allowed input.
5. Add literals such as `/` or `-` when needed.
6. Enter the value according to the defined mask.

## 💡 Example

`MaskedTextBox1.Mask = "00/00/0000";`

### ➡ Result

The control appears as:

\_\_\_/\_\_\_/\_\_\_\_

Example input:

10/09/2026

## Note

### ◆ Common mask characters:

-  `0` → required digit.
-  `9` → optional digit or space.
-  `L` → required letter.
-  `?` → optional letter.
-  `A` → required alphanumeric character.
-  `&` → required character.
-  `C` → optional character.

### ◆ Case conversion:

-  `<` → converts following characters to lowercase.
-  `>` → converts following characters to uppercase.
-  `|` → disables the previous case conversion.

### ◆ Escape character:

-  `\` → treats the following mask character as a literal.



# Comparison

## Required

- ✓ Must contain a valid character.
- ✓ Examples: `0`, `L`, `&`, `A`.

## Optional

- ✓ Can be left blank.
- ✓ Examples: `9`, `?`, `C`, `a`.

## ⊘ Warning

- ⌘ A mask controls the input format but does not guarantee that the value is logically valid.
- ⌘ For example, `00/00/0000` accepts a date-like format but does not verify whether the date is a real calendar date.

# ✦ Summary

✓ MaskedTextBox restricts input using a custom mask.

- `Mask` → defines the input format.
- `0` → required digit.
- `9` → optional digit.
- `L` → required letter.
- `A` → required alphanumeric character.
- `<` / `>` → control letter case.
- `\\` → escapes a mask character.
- Other characters → fixed literals.

Thank you 😊